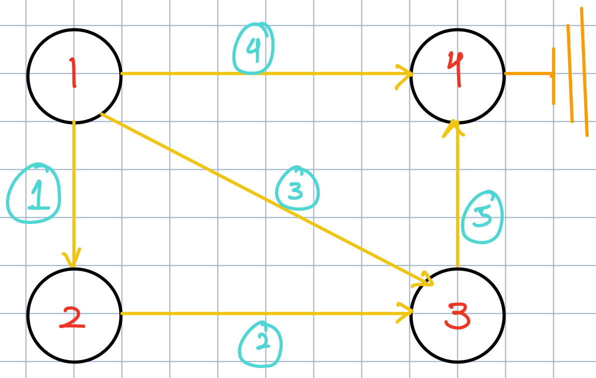


$n = 4$ nodes
$m = 5$ edges

Incidence matrix:

$$A = \begin{array}{c} \\ \\ \end{array}\begin{array}{cc} & \begin{array}{cccc} \text{node 1} & \text{node 2} & \text{node 3} & \text{node 4} \end{array} \\ & \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \end{array} \begin{array}{l} \text{edge 1} \\ \text{edge 2} \\ \text{edge 3} \\ \text{edge 4} \\ \text{edge 5} \end{array}$$

These are dependent rows & thus they form a row

Null space: $[x_1 \; x_2 \; x_3 \; x_4]$ Potentials at the node

$$Ax = 0$$

$$\underbrace{\begin{bmatrix} x_2 - x_1 \\ x_3 - x_2 \\ x_3 - x_1 \\ x_4 - x_1 \\ x_4 - x_3 \end{bmatrix}}_{\text{Potential differences}} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$\xrightarrow[\text{Ohm's law}]{C}$ Currents y_1, y_2, y_3, y_4, y_5 on edges $\rightarrow$ $A^T y = 0$ Kirchoff's CL

$$x = c \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

↳ If all potentials are same then no current will flow.

↳ So, we fix one potential & the matrix becomes

$$A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

This matrix is linearily independent

↳ Three pivot columns, so only free variable & we ground it

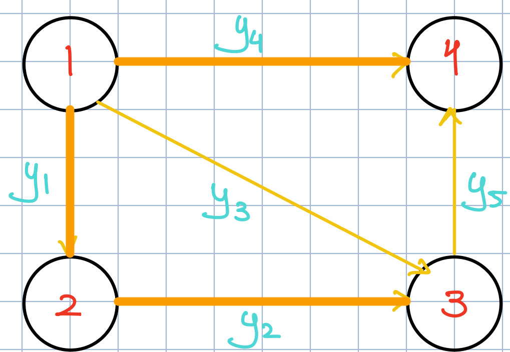
Null space A^T:

$$A^Ty = 0$$

$$\begin{array}{ccccc} \text{pivot} & \text{pivot} & & \text{pivot} & \\ \end{array}$$
$$\begin{bmatrix} -1 & 0 & -1 & -1 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix}$$

$$-y_1 - y_3 - y_4 = 0$$
$$y_1 - y_2 = 0$$
$$y_2 + y_3 - y_5 = 0$$
$$y_4 + y_5 = 0$$

net flow is 0



↳ Thick edges is same graph with no loops
↳ Called as Tree

Nullity = cols of A − rank A

$$\#\text{loops} = \#\text{edges} - (\#\text{nodes} - \underbrace{1}_{\text{grounded node}})$$

$$\#\text{nodes} - \#\text{edges} + \#\text{loops} = 1$$

Euler's formula

Basis for Null A^T:

$$\begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}$$

Any loop satisfies the law

$$e = Ax$$

$$A^T y = f$$

$$y = Ce$$

$$\underbrace{A^T \underbrace{C \underbrace{A}_{P.D}}_{\substack{\text{physical}\\ \text{constants}\\ (Y)}} x}_{f} = f$$